Questions on array (1D)

1. Write a program to take 20 integers in an array D[]. Print sum of all even and product of all odd numbers.

2. Write a program to take 15 integers in an array and print count of positive and count of negative numbers

3. Write a program to take n integers from user and find the smallest and the greatest from the array

4. Write a program to store marks of 50 students in a single subscripted variable and print number of students falling in the following categories in two columns

Marks	Number of students
1-30	--
31-50	--
51-70	--
71-85	--
86-100	--

5. Write a program to take a single dimensional array of N real numbers from user. Print only prime integers from the array

6. Write a program to create a single dimensional array of 20 integers. Print count of only double-digit integers and count of triple digit integers from user

7. Write a program to create a single dimensional array if N real numbers. Print the rounded value of those
   Real numbers whose fractional part is greater than or equal to 5

8. Write a program to input a single subscripted variable AR[] of 50 integers. Remove all duplicate elements from the array so that each element of the array should appear only once. Print the original and the modified array

9. Write a program to take two arrays of N size (A[] and B[]), check and print if both are identical or not

10. Write a program to take a single dimensional array of N integers. Print only those integers from the array which are Armstrong Numbers.

11. Write a program to take a single dimensional array of n integers. Print only integers from the array which are palindrome numbers

12. Write a program to take N integers in an array. Print sum of negative, positive even, positive odd numbers

13. Write a program to take N integers in an array. Print sum of all integers present at odd indexes and product of all odd elements at even indexes

14. Write a program to take N integers in an array. Make another array and arrange elements from the first array such that even elements appear from left to right and odd elements from right to left

15. Design a method void show(int num[ ], int d) containing integer elements. Count the occurrence of digit d in num[] array and print it. Call the function in main()

16. Design a method void Frequency(int arr[]) to count and print frequency of each single digit number in arr

17. Write a program to take N integers in an array. Take an index from user from user. Print table of the number present at that index

18. The annual examination results of 50 students in a class is tabulated as follows:

    Roll no Subject A Subject B Subject C

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    Write a program to read data and display the following:
    a) Average mark obtained by each student
    b) Print roll numbers and average marks of the students whose average is more than 80
    c) Print roll numbers and average marks of the students whose average is below 40
    d) Print roll of the student with highest average mark

19. Write a program to create a function void findGCD(int N[]) which takes a single dimensional array with
    8 integers. Read every two integers and print their GCD/HCF

20. Design functions :
    int factorial(int x) to calculate and return the factorial of number stored in x
    void isSpecial(int arr[]) : Print only special numbers from the array using the function factorial(x)
    Special Number: sum of factorial of each digit is equal to the number

21. Write a program to take N integers in an array and sort them using bubble sort in ascending order. Print the original and the sorted array

22. Write a program to take N integers in an array and sort them using selection sort in ascending order. Print the original and the sorted array

23. Write a program to accept and store Roll number and Total marks of 'N' students in the two single dimensional arrays, roll[] and marks[ ]. Arrange the array roll[ ] in Ascending order along with marks[ ] using Selection Sort technique. Print the sorted arrays in two columns.

24. Design a function void Sort(int total[ ], double per[ ]) to arrange the percentage stored in the argument per[] in Descending order along with total[ ] using Bubble Sort technique. Print the sorted arrays in two columns.

25. Write a program to create three arrays roll[ ] (long integer), total[ ] (double type) and per[ ] (double type) to input and store roll nos., total marks and percentage of 40 students in a class. Generate a merit list on the basis of percentage and print it along with roll[ ], total[ ] and per[ ].
[Hint: Merit list means arrangement from higher percentage to lower percentage].

26. Write a program to input 20 integers in an array and an integer num, from the keyboard. Search the integer 'num' from the array using Linear search technique and print the index of the num if found in the array otherwise print a message "Number not found".

27. Design a function int Search(int num[], int D) to search the value in the argument 'D' from the array
num[ ] using Linear search technique and return the index if 'D' found in the array otherwise return -1.

28. Write a program to define a class Bsearch to initialize an array val[ ] with 15 integers in Ascending order. Input an integer num from the keyboard and check if num is present in the sorted array or not. Print the index or subscript, if num is found otherwise print an appropriate message. Use Binary search method for searching.

29. Design a function void Search(int num[ ], int V) to sort the array num[ ] in Descending order using any sorting technique and search the value in the argument 'V' from the sorted array num[] using Binary search method and print the index along with the number if 'V' is found in the array otherwise print "Not Found".

30. Design a function int BinSearch(int Roll[ ], int Marks[], int rn), where the argument Roll[ ] is already arrange in Ascending order along with the array Marks[ ]. Search the value in the argument 'rn' from the array Roll[ ] using Binary search method and return the index if 'rn' is found in the array otherwise print "Search unsuccessful" and return -1.

31. Design a function void BinarySearch(double total[ ], double percentage[ ], double pr), where the argument percentage[] is already sorted in Descending order along with the array total[ ]. Search the value in the argument 'pr' from the array percentage[] using Binary search method and print the total and percentage along with index if pr' is found in the array otherwise print "Search unsuccessful".

32. Write a program to perform binary search on the list of integers given below (that are arranged in Ascending order), to search for an element input by user, if it is found display the element along with its position, otherwise display the message "Search element not found".
    5, 7, 9, 11, 15, 20, 30, 45, 89, 97

33. Write a program to store 6 elements in an array P, and 4 elements in array Q and produce a third array R, containing all the elements of arrays P and Q. Display the resultant array.
34. Write a program to input and store weight of ten people. Sort and display them in Descending order using the Selection sort technique.
35. Write a program to input ten (10) integer elements in an array and sort them in Descending order using the Bubble sort technique.

36. Write a program to accept the year of graduation from school as integers from the user. Using the Binary Search technique on the sorted array of integers, output the message "Record exists", if the year input is located in the array. If not, output the message "Record does not exist".
    [1982, 1987, 1993, 1996, 1999, 2003, 2006, 2007, 2009, 2010]

37. Write a program to input and store roll numbers and marks in 3 subjects of n number of students in four single dimensional arrays and display the remark based on average marks as given below:

Average Marks	**Remarks**
85-100	EXCELLENT
75-84	DISTINCTION
60-74	FIRST CLASS
40-59	PASS
Less Than 40	FAIL

38. Write a program to initialize the following array [2, 5, 4, 1, 6, 8, 5]
    Insert a user given number m at user given index k

39. Write a program to initialize the following array [2, 5, 4, 1, 6, 8, 5]
    Delete element stored in user given index k

40. Write a program to initialize the following array [2, 5, 4, 1, 6, 8, 5]
    Delete a user given element p

41. Write a program to take N elements in an array. Create another array to store only elements which appear more than once (duplicate elements)

42. Write a program to take 100 elements in an array and find Mean, Median and Mode of the elements

43. Take 50 integers in an array from user. Sort first half in ascending order and last half in descending order

44. Take an array of size N, reverse it, store in another array

45. Take an array of size N, reverse it in its own place without using another array

46. Write a program to take N integers in an array. Print all pairs of numbers which are **twin primes** (i.e., both are prime and differ by 2) and exist in the array.